Amiri Hayes

Machine Learning Engineer
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EDUCATION:

NJIT | New Jersey Institute of Technology Sep 2023 - Dec 2026
MS Artificial Intelligence, BS Applied Mathematics (with Honors) [Full Ride Merit Scholarship]

Select Courses: Deep Learning, Natural Language Processing, Stochastic Simulation, Reinforcement Learning, Computer Vision, Numerical & Statistical Methods, Linear Algebra, Real Analysis I & II, AI for Robotics

RCSJ | Rowan College of South Jersey Triple Major: AS Physics, AS Mathematics, AS Computer Science

RECENT PAPERS:

- *Amiri Hayes*, Belinda Li, Jacob Andreas. **Explaining Attention with Program Synthesis.** NeurIPS 2026. (under review) Project: Established precedence for replacing LLM attention heads with a dataset of generalizable executable programs, which provides full insight into logic at the attention stage while maintaining model performance.
- *Amiri Hayes*, Kristina Wicke. **Systematic Construction of Text-Attributed Graphs.** (in preparation) Project: Developed system to train decision trees based on text-attributed graph construction choices. We next analyze correlations between our empirically validated trees to GraphRAG quality across question-answering contexts.

EXPERIENCE:

linkedin.com/in/amirihayes – github.com/AmiriHayes

MIT CSAIL | LINGO, Language & Intelligence Group Cambridge, MA | MIT
Fall 2026 - Spring 2027 · Pre-Doctoral Researcher
- Incoming year-long lab member focused on converting LLM activations to Python for mechanistic interpretability

Summer 2025 · Machine Learning Researcher, MSRP Intern
- Engineered Python interpretability library "ViewLLM" using program synthesis to classify LLM attention heads
- Built distributed Python & SLURM pipelines on CSAIL H100 clusters for scalable experiments and data flow
- Classified ~1/2 of BERT activations + automated over 300 hypothesis functions, advancing head interpretability

IBM Research | Thomas J. Watson Research Center Yorktown Heights, NY | IBM
Summer 2026 · Machine Learning Researcher, GEM Fellow Awardee
- Developing diffusion generative models to solve combinatorial optimization problems over discrete solution spaces
- Training supervised, unsupervised, and quantum solvers applied to Max3SAT & MaxCut hypergraph instances
- Conducting experiments on industry-scale compute; targeting preprint submission at NeurIPS 2026 workshop

Brookhaven National Laboratory | Environmental Sciences & Technologies Upton, NY | BNL
Spring 2026 · Computational Sciences Researcher, SULI Intern
- Engineered statistical framework to identify cold fronts from ~100GB of climate data with 90%+ inference accuracy
- Developed spatiotemporal inference models to extrapolate event lifecycles & identify atmospheric transition timing
- Conducted correlation analysis of variables; saved results and loaded data via GCP to optimize loading performance

UCLA | Institute for Pure & Applied Mathematics Los Angeles, CA | UCLA
Summer 2024 · Machine Learning Researcher, RIPS Intern
- Led team of 3 undergrads as project manager developing generative models for epidemiological network simulation
- Implemented VAE, GAN, and score-based algorithms w/ parallel computing to generate large social graph datasets
- Generated synthetic 100-person social networks with vaccination statistics, developed a network fusion technique

RELEVANT SKILLS:

Specializations: Machine Learning, Deep Learning, Data Engineering, AI Safety, Language Models
Certifications: Google Data Analytics, Cornell Break Through Tech AI Program Cohort 2026
Technical: Python, PyTorch, JavaScript, SQL, C++, AWS, Google Cloud, MongoDB, ReactJS, HuggingFace